

BECOMING A DATA HEAD

CHAPTER 12: DEEP LEARNING

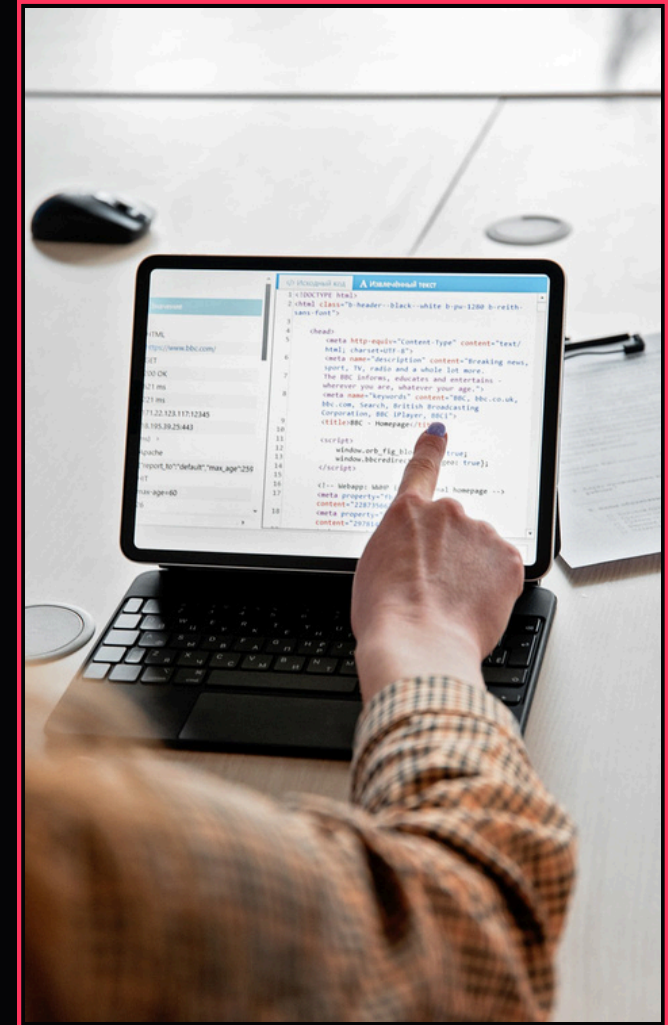
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Naiafian
9/17/2026**

DEEP LEARNING: THE BIG PICTURE

Powerful, exciting – but not magic.

Image recognition
Speech and translation
Autonomous driving
Medical imaging
High-stakes decisions

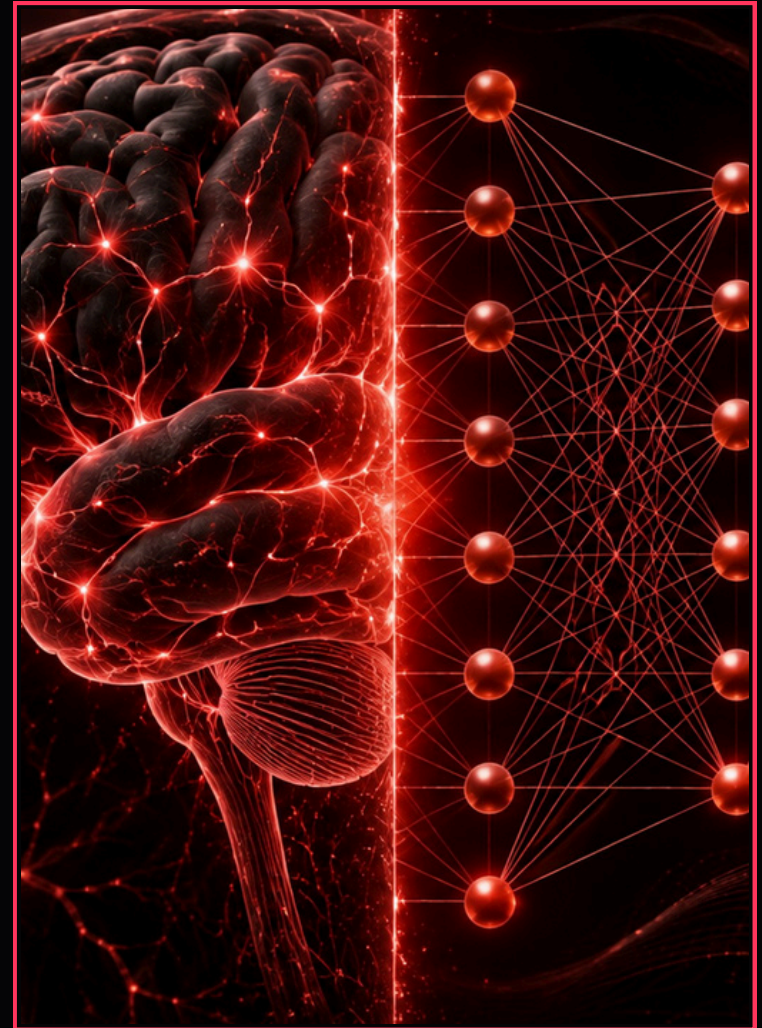
Core message: deep learning is impressive, but it still depends on data and judgment.



NEURAL NETWORKS ARE NOT BRAINS

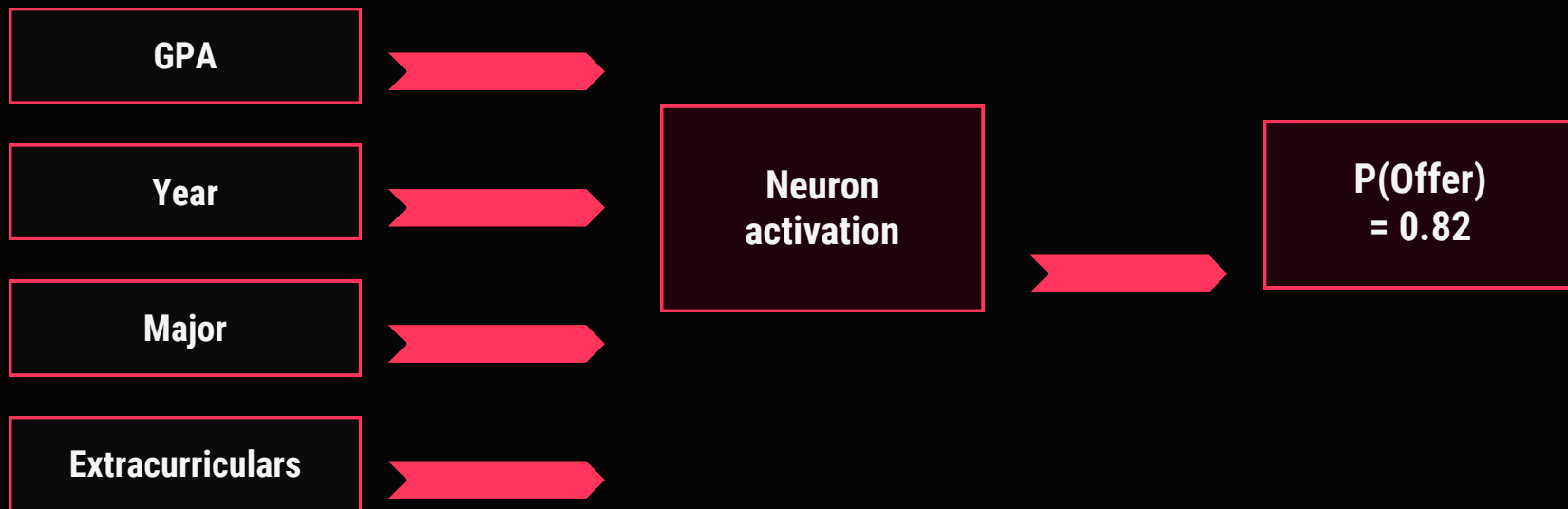
Brain-inspired is not brain-equivalent.

The analogy is useful at first.
But it can create hype.
Modern success comes from data, compute,
and research.
A neural network is still a giant math model.



THE SIMPLEST NEURAL NETWORK

Four applicant inputs become one probability.



This is why Chapter 12 connects directly to logistic regression.

INSIDE A NEURON

Weighted sum + activation function.

$$z = w_1x_1 + w_2x_2 + \dots + b$$

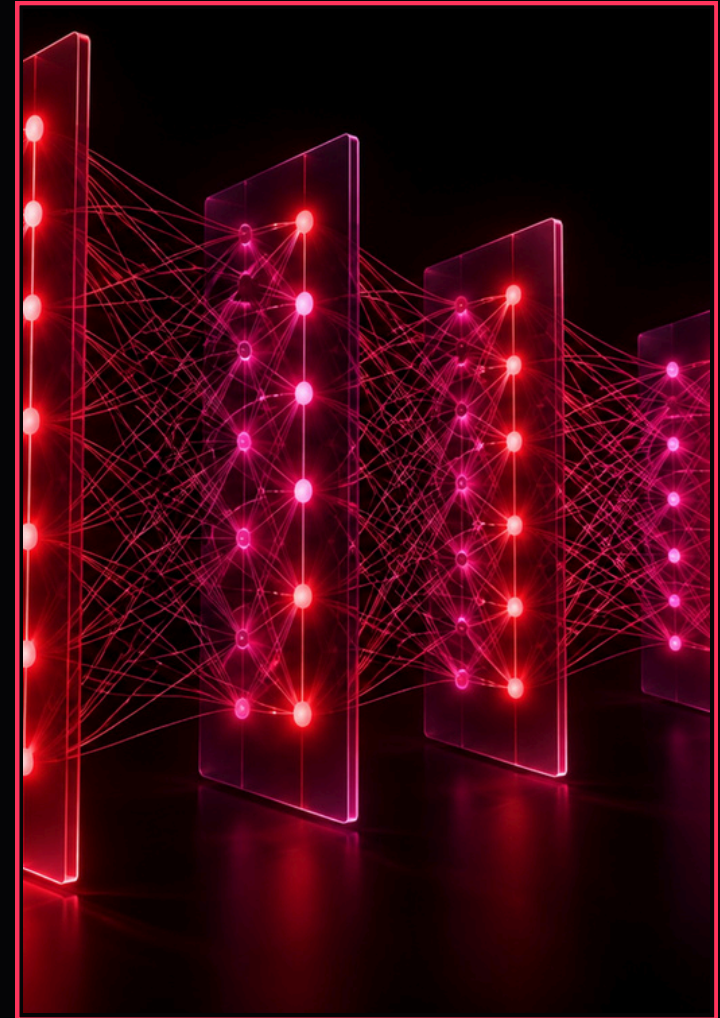


Training means finding useful values for the weights and bias.

WHAT DOES THE NETWORK LEARN?

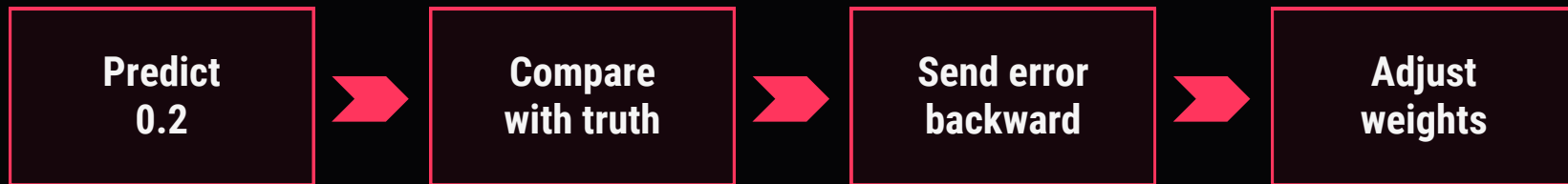
It learns weights and biases.

Weights decide which inputs matter more.
Bias shifts the prediction up or down.
Good parameters make predictions close to real labels.
Bad parameters create large errors.



BACKPROPAGATION

Learn from the error.



Prediction → error → weight update → better prediction.

WHY ADD HIDDEN LAYERS?

They create internal representations.

GPA + Major + EC



Achievement

Year + EC

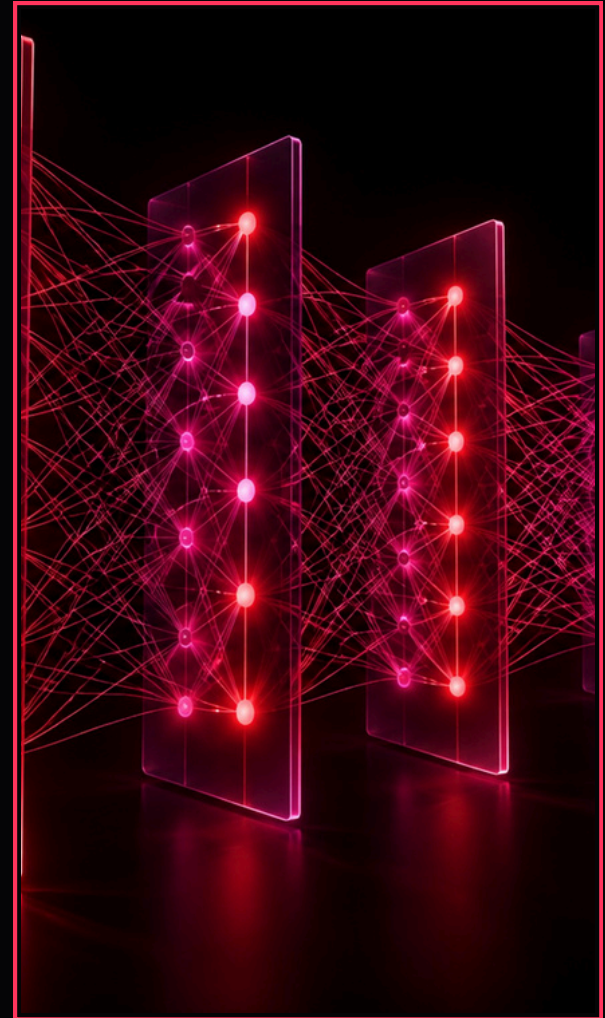


Experience

Major + Skill



Skillset



REPRESENTATION LEARNING

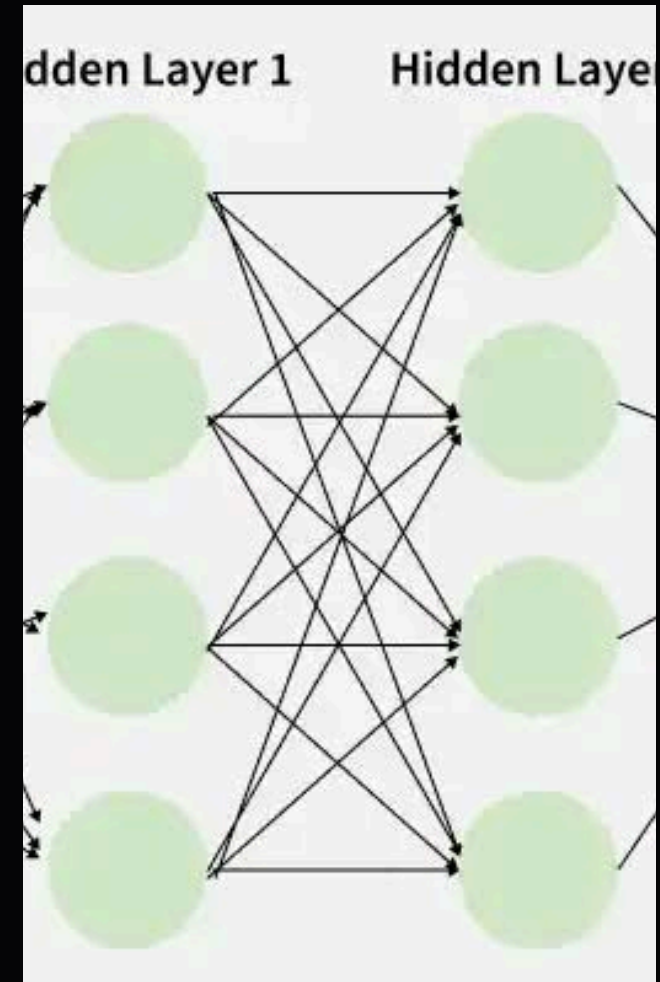
The real superpower of deep learning.

Traditional ML: humans create useful features.

Deep learning: hidden layers learn useful features.

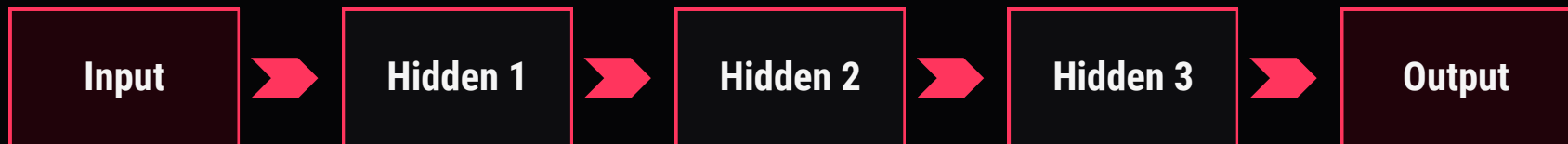
This is why it works well on images, audio, video, and text.

It reduces manual feature engineering – not human responsibility.



WHAT MAKES IT DEEP?

Multiple hidden layers stacked together.

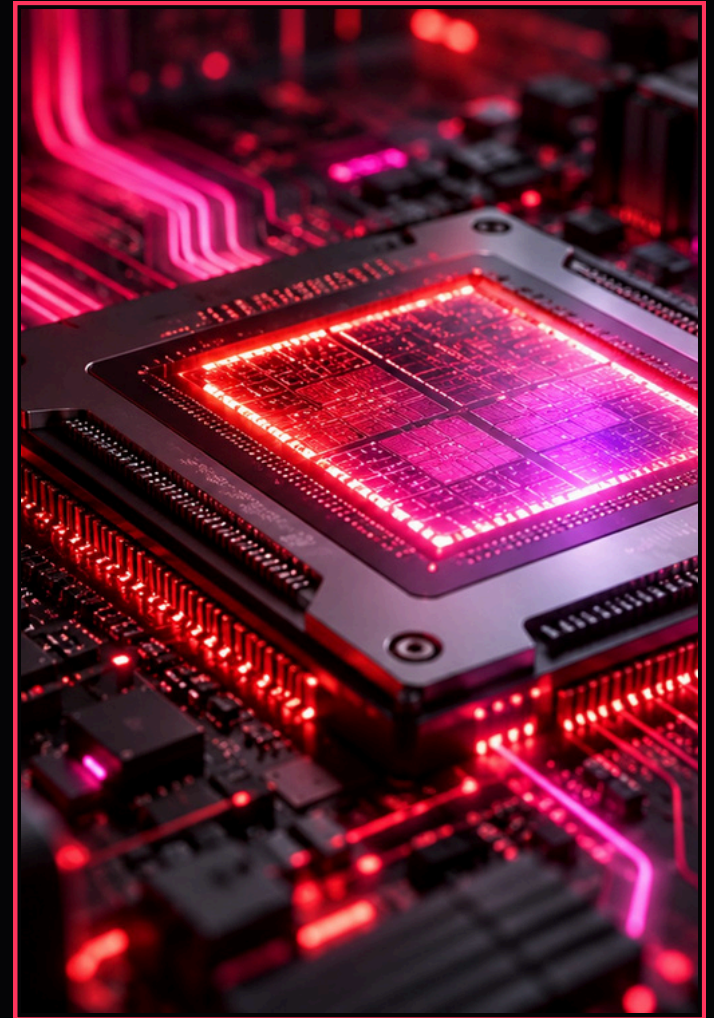


Depth = repeated layers of representation.

NEURAL NETWORKS ARE OLD

The boom is new; the idea is not.

Early neural networks date back decades.
Deep learning struggled when data and
compute were limited.
The modern breakthrough came from better
conditions.
Don't confuse old ideas with weak ideas.



WHY DEEP LEARNING TOOK OFF

Three ingredients came together.

**Big
Data**

+

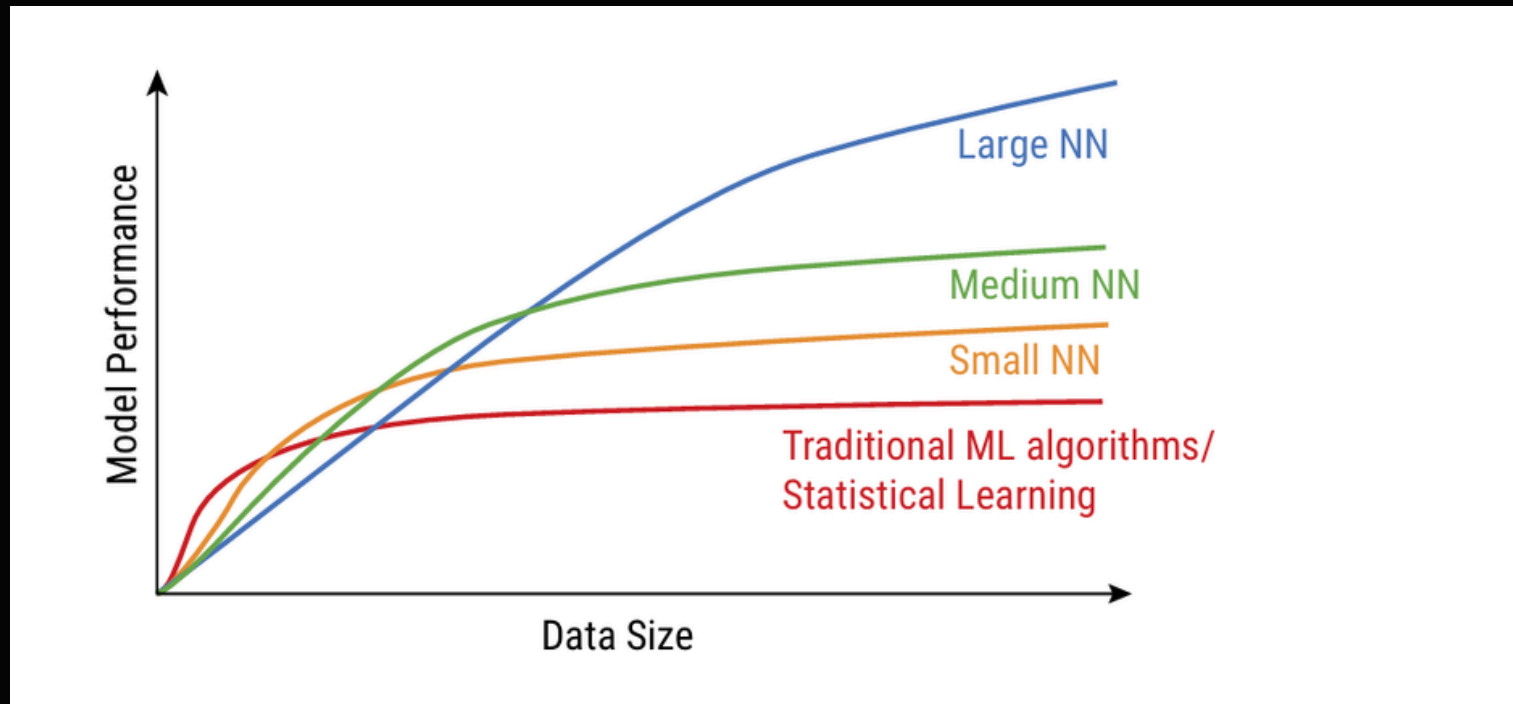
**Better
Algorithms**

+

GPUs

The revolution was about scale: more data, more compute, better optimization.

MORE DATA CAN HELP

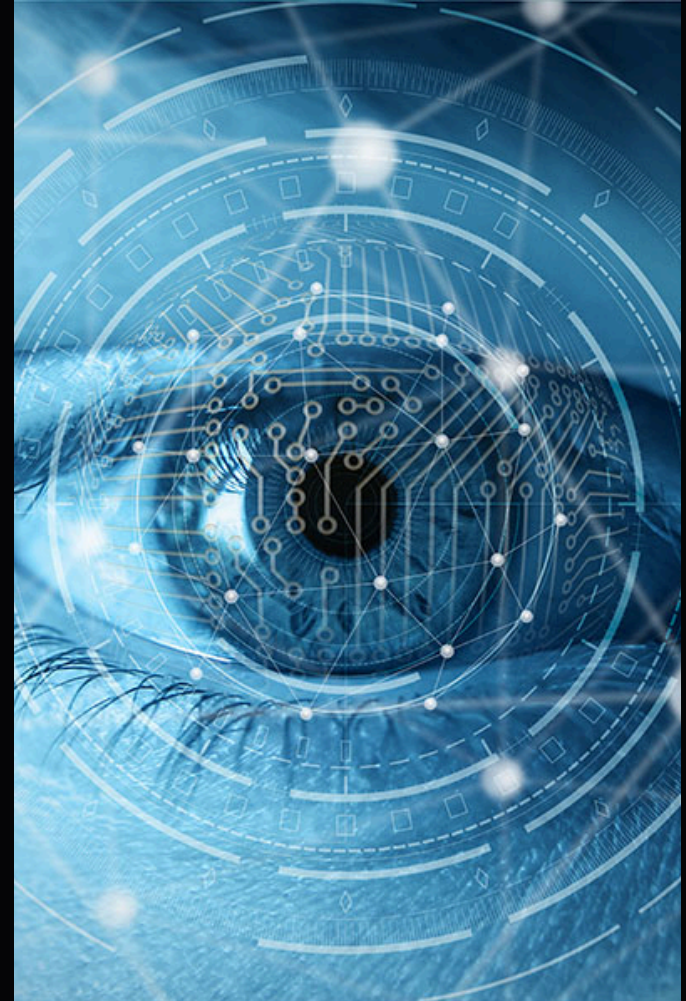


Important caveat: more data only helps if the data contains real signal.

HOW COMPUTERS SEE IMAGES

Humans see a digit; computers see numbers.

A grayscale image is a matrix of pixel values.
An 8×8 image becomes 64 numbers.
The network learns patterns from those numbers.
It does not “see” the way we see.



COLOR IMAGES GET HUGE

Every pixel has three channels.

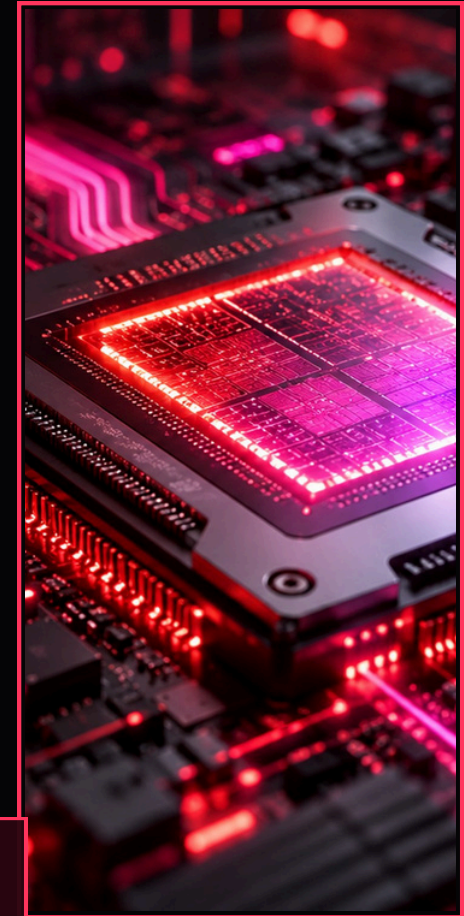
Red

Green

Blue

10 MP image $\approx$ 30 million values

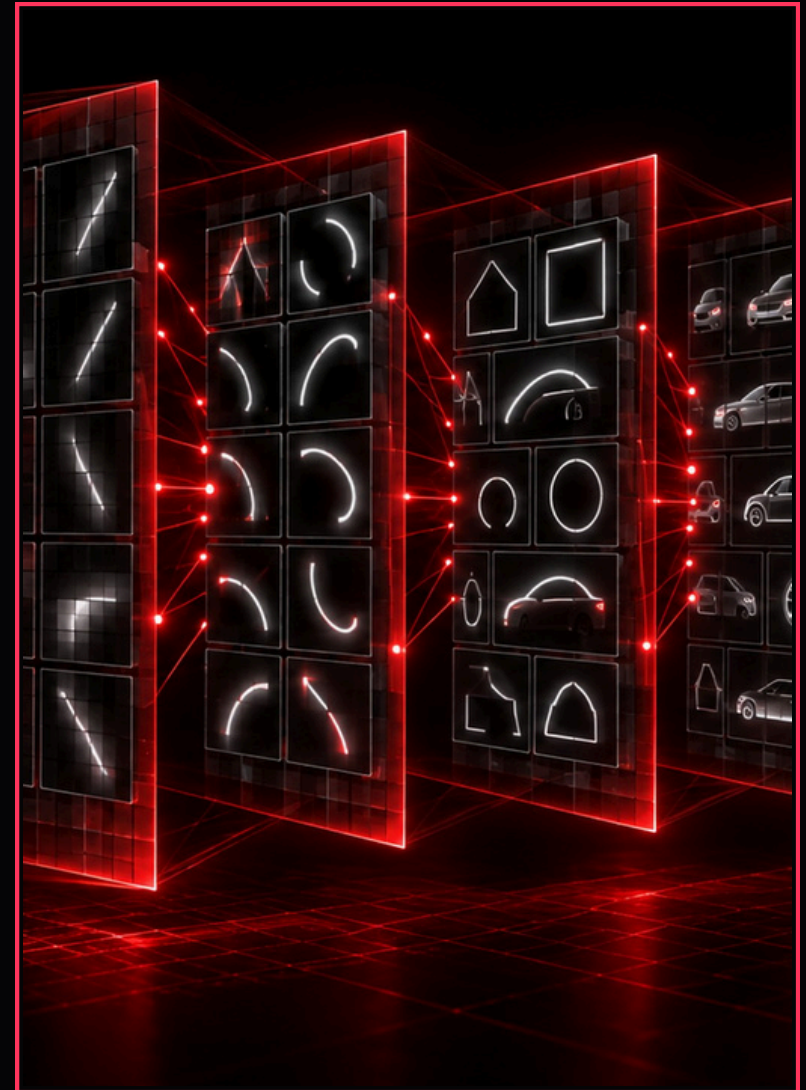
A fully connected network would become computationally ridiculous.



CONVOLUTIONAL NEURAL NETWORKS

Like moving magnifying glasses across an image.

Filters scan local regions.
They detect edges, corners, curves, and textures.
Later layers combine simple patterns into objects.
This is why CNNs are powerful for images.



POOLING

Keep the important signal, reduce the size.



Convolution finds local patterns. Pooling summarizes them.

WORD ORDER MATTERS

Same word. Different context.

At breakfast, I like to drink orange _____

My cousin in California lives in Orange _____

juice? county?

RECURRENT NEURAL NETWORKS

A simple form of memory for sequences.

Current input depends on previous information.
Useful for language and sequences.
The network can remember earlier context.
This fixes part of the Bag-of-Words limitation.



FIRST QUESTION: DO WE HAVE DATA?

Deep learning needs labeled examples.

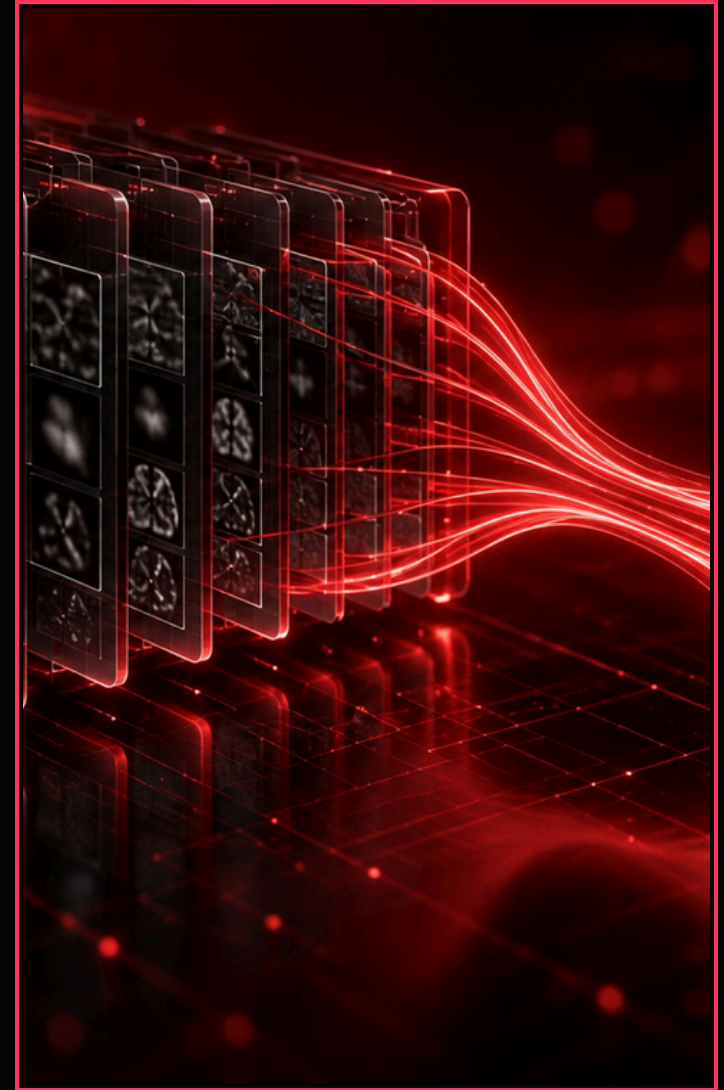
Not just data — labeled data.
Labels must match the specific task.
Poor labels create poor models.
No model can learn a relationship that is not in the data.



TRANSFER LEARNING

Do not always start from zero.

Use a pretrained model.
Keep generic early features like edges and shapes.
Replace or retrain final layers.
Useful when labeled data is limited – but not absent.



DEEP LEARNING IS NOT ALWAYS BEST

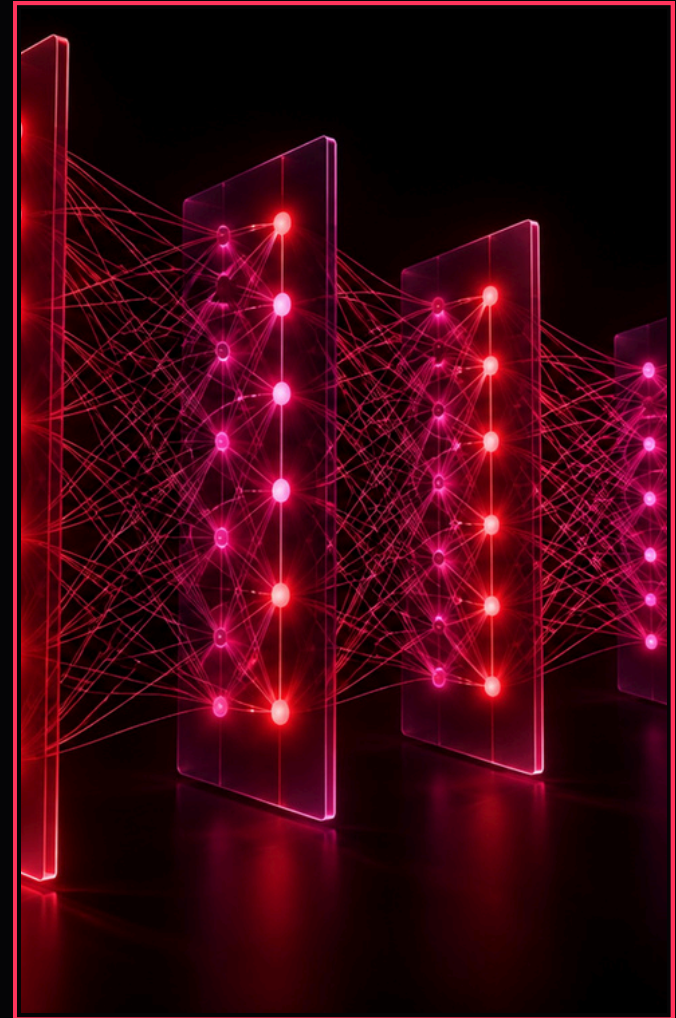
Choose the model for the data — not for the hype.

Strong on perceptual data: images, text, audio, video.

Often not best for small structured tables.

Tree-based models can be better on tabular data.

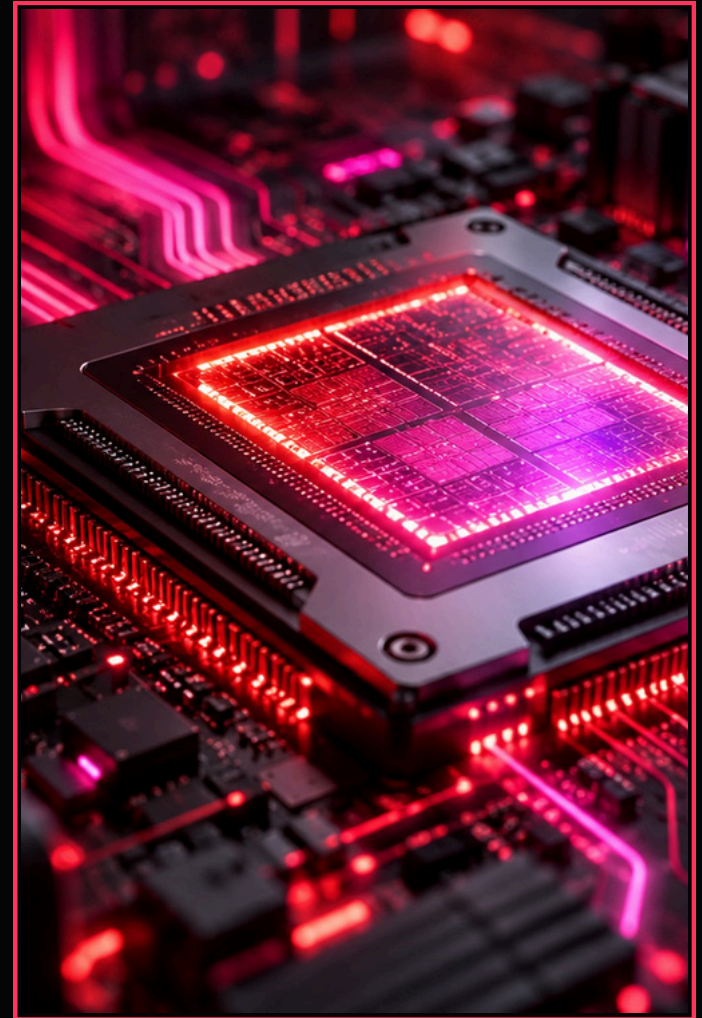
If all models fail, investigate the data first.



ARCHITECTURE CHOICES

A network is not “click train and magic.”

How many layers?
How many neurons?
Which activation functions?
How to avoid overfitting?
How to validate performance?



THE BLACK BOX PROBLEM

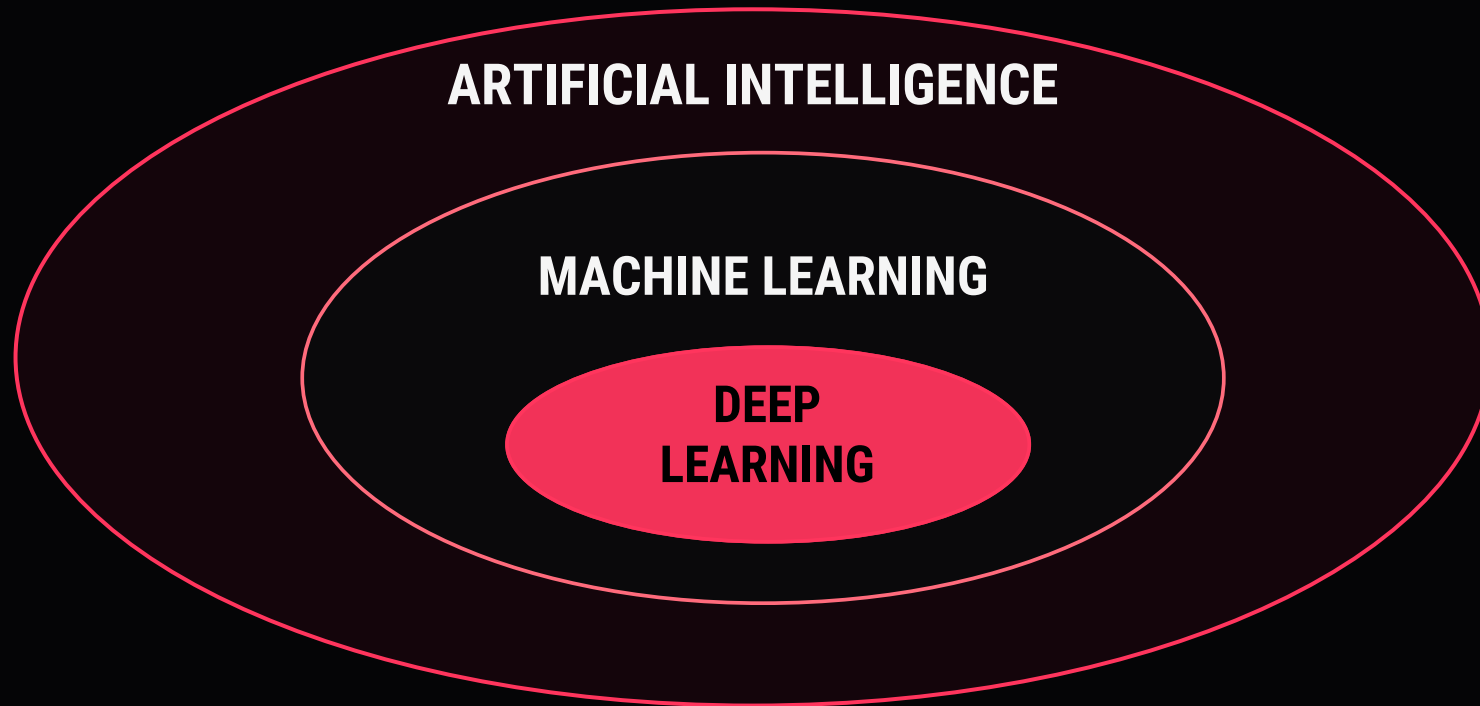
More complexity usually means less interpretability.

Millions of parameters are hard to explain.
Hidden representations may not have human meanings.
Good predictions can still be difficult to justify.
This matters more in high-stakes decisions.



AI, MACHINE LEARNING, DEEP LEARNING

The terms are related, but not identical.



Most “AI” in practice is machine learning; deep learning is one powerful subset.

BIG TECH'S HIDDEN ADVANTAGE

The glamorous part is the model. The moat is often the data.

Years of labeled images, clicks, speech, text, and interactions.

Users often help create labels without noticing.

Better data can matter more than a flashier algorithm.

Data collection is part of the system.



AI ethics, surveillance and ...



YOU ARE
BEING
WATCHED

THE DATA HEAD QUESTION

End with responsibility, not mystique.

WHO DOES THIS RESULT AFFECT?

AM I OK WITH THAT?

Deep learning is powerful. It is not neutral. Humans still stand behind the work.

THANK YOU

QUESTIONS & DISCUSSION

